1. Consider the following regression equation:

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. What does β_1 imply?

a. β_1 measures the ceteris paribus effect of x_1 on x_2.

b. β_1 measures the ceteris paribus effect of y on x_1.

c. β_1 measures the ceteris paribus effect of x_1 on y.

d. β_1 measures the ceteris paribus effect of x_1 on u.

2. Which of the following is true of R^2?

a. R^2 is also called the standard error of regression.

b. A low R^2 indicates that the Ordinary Least Squares line fits the data well.

c. R^2 usually decreases with an increase in the number of independent variables in a regression.

d. R^2 shows what percentage of the total variation in the dependent variable, Y, is explained by the explanatory variables.

3. Consider a regression model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. The value of R^2 always _____.

a. lies below 0

b. lies above 1

c. lies between 0 and 1

d. lies between 1 and 1.5

4. If an independent variable in a multiple linear regression model is an exact linear combination of other independent variables, the model suffers from the problem of _____.

a. perfect collinearity

b. homoskedasticity

c. heteroskedasticty

d. omitted variable bias

5. Exclusion of a relevant variable from a multiple linear regression model leads to the problem of _____.

a. misspecification of the model

b. multicollinearity

c. perfect collinearity

d. homoskedasticity

6. Suppose the variable x_2 has been omitted from the following regression equation, $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$. $\widetilde{\beta}_1$ is the estimator obtained when x_2 is omitted from the equation. If $E(\widetilde{\beta}_1) > \beta_1$, $\widetilde{\beta}_1$ is said to _____.

a. have an upward bias

b. have a downward bias

c. be unbiased

d. be biased toward zero

7. High (but not perfect) correlation between two or more independent variables is called _____.

a. heteroskedasticty

b. homoskedasticty

c. multicollinearity

d. micronumerosity

8. Find the degrees of freedom in a regression model that has 10 observations and 7 independent variables in a multiple regression model with an intercept.

a. 17

b. 2

c. 3

d. 4

9. The Gauss-Markov theorem will not hold if _____.

a. the error term has the same variance given any values of the explanatory variables

b. the error term has an expected value of zero given any values of the independent variables

c. the independent variables have exact linear relationships among them

d. the regression model relies on the method of random sampling for collection of data

10. The key assumption for the general multiple regression model is that all factors in the unobserved error term be correlated with the explanatory variables.

11. The coefficient of determination (R^2) decreases when an independent variable is added to a multiple regression model.

a. True

b. False

12. An explanatory variable is said to be exogenous if it is correlated with the error term.

a. True

b. False

13. A larger error variance makes it difficult to estimate the partial effect of any of the independent variables on the dependent variable.

a. True

b. False

14. When one randomly samples from a population, the total sample variation in x_j decreases without bound as the sample size increases.

a.True

b.False